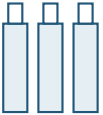
Measured cohort



139 independent bottles

11 constituents

Spectral choices

7 non-empty masks

Training-only rule

One shared mask for the full panel

Nested validation
4 model families

Observed results

U + R

2.8-6.5% lower panel MSE vs U+R+N

80-92% retained in bootstrap refits

Target-level exceptions remain